Exhibit 3.7 represents investment risk (aka market risk) as distinct from sequence-of-returns risk. Average returns over different thirty-year rolling periods fluctuate. They range from about 3 percent for the period starting in 1952 to roughly 7 percent for the period starting in 1982. Note that rolling thirty-year returns never fell too close to the 1.3 percent compounded real return that would trigger a maximum sustainable spending rate of 4 percent. We can think about this volatility for compounded returns over a thirty-year period as the general investment risk facing retirees. However, sequence-of-returns risk has not been factored in yet, so this is not the whole story.

Exhibit 3.8 demonstrates the specific impact of sequence risk compared to more general investment risk by showing historical withdrawal rates for two sets of return assumptions. The lighter curve shows withdrawal rates assuming that the fixed compounded returns shown in the previous exhibit were earned without any market volatility within each thirty-year period. These numbers represent a simple PMT calculation: with withdrawals taken at the start of the year and a fixed compounded portfolio return for the next thirty years, what spending rate would deplete the portfolio in the thirtieth year?

Because the average investment returns fluctuate, the hypothetical sustainable withdrawal rates also fluctuate. Nonetheless, they remain within a range of about 2.5 percentage points. They do not fall below 5 percent, and they are a little above 7.5 percent at their highest. These would be sustainable withdrawal rates if the average compounded returns over thirty years were all that mattered.

The darker curve in the exhibit represents the actual sustainable withdrawal rates over those thirty-year periods, as previously shown in Exhibit 3.5. What matters for calculating the actual withdrawal rates is not just an average portfolio return, but also the specific sequence of returns. Actual withdrawal rates are more volatile and include greater downside after accounting for sequence risk—the withdrawal rate fell as low as 4 percent for a 1966 retiree. On the other hand, greater upside potential exists as well—a 1982 retiree could have used a 9.8 percent withdrawal rate. Retirees can indeed be adversely affected in ways that dig deeper than the investment risk associated with the average return experienced over a long retirement.